

All Descendants (Query 01)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
F40	<i>Q_{desc}</i>	Root	9.6	1.4	1.4	6.7x	6.9x	3.0	1.9	1.9	1.6x	1.6x	2.0	2.0	2.3	0.99x	0.88x
F40	<i>Q_{desc}</i>	Leaf parent H	8.8	1.4	1.4	6.3x	6.4x	2.7	1.4	1.3	1.9x	2.1x	1.3	1.6	1.5	0.86x	0.87x
F40	<i>Q_{desc}</i>	Leaf parent	8.9	1.4	1.4	6.4x	6.4x	2.7	1.9	1.7	1.4x	1.6x	1.3	1.5	1.7	0.86x	0.78x
F1K	<i>Q_{desc}</i>	Root	441.6	2.3	2.3	190x	190x	6.3	3.1	2.5	2.1x	2.5x	29.5	28.5	41.5	1.0x	0.71x
F1K	<i>Q_{desc}</i>	Leaf parent H	13.3	2.3	2.3	5.9x	5.9x	4.2	1.9	2.2	2.2x	1.9x	1.9	2.8	2.3	0.68x	0.85x
F1K	<i>Q_{desc}</i>	Leaf parent	13.0	1.9	1.8	6.7x	7.1x	3.6	2.0	2.0	1.7x	1.8x	1.8	1.7	1.9	1.0x	0.95x
NT10	<i>Q_{desc}</i>	Root	9.3	1.3	1.3	7.0x	6.9x	2.8	2.0	1.7	1.4x	1.6x	2.2	2.1	2.3	1.0x	0.96x
NT10	<i>Q_{desc}</i>	Leaf parent H/L	9.3	1.3	1.3	7.0x	6.9x	2.6	1.7	1.5	1.5x	1.7x	1.4	1.7	1.6	0.83x	0.88x
NT100	<i>Q_{desc}</i>	Root	20.1	1.6	1.5	13x	13x	3.9	1.7	1.8	2.2x	2.2x	11.5	13.3	13.7	0.86x	0.84x
NT100	<i>Q_{desc}</i>	Leaf parent H	9.9	1.5	1.4	6.8x	7.0x	3.1	1.8	1.6	1.7x	1.9x	2.2	2.4	2.3	0.91x	0.94x
NT100	<i>Q_{desc}</i>	Leaf parent	8.8	1.4	1.4	6.1x	6.2x	2.9	1.5	1.7	1.9x	1.8x	1.5	1.6	1.8	0.95x	0.83x
NT1K	<i>Q_{desc}</i>	Root	1,114.5	2.8	2.8	400x	390x	10.1	2.0	2.0	5.2x	4.9x	57.7	46.0	47.6	1.3x	1.2x
NT1K	<i>Q_{desc}</i>	Leaf parent H	15.5	2.3	2.3	6.7x	6.8x	2.9	1.5	1.7	1.9x	1.7x	1.3	1.4	1.5	0.93x	0.90x
NT1K	<i>Q_{desc}</i>	Leaf parent	12.5	2.3	2.3	5.4x	5.5x	3.8	1.7	1.8	2.2x	2.1x	1.2	1.2	1.3	0.99x	0.90x
NT10K	<i>Q_{desc}</i>	Root	107,135.3	15.6	15.6	6.9Kx	6.9Kx	101.3	10.7	9.2	9.5x	11x	416.5	488.0	533.7	0.85x	0.78x
NT10K	<i>Q_{desc}</i>	Leaf parent H	269.0	11.5	9.7	23x	28x	2.7	4.8	2.5	0.56x	1.1x	1.4	1.4	1.4	1.0x	1.0x
NT10K	<i>Q_{desc}</i>	Leaf parent	245.4	11.6	9.8	21x	25x	3.0	5.9	2.2	0.51x	1.4x	1.2	1.8	1.9	0.67x	0.65x
NT100K	<i>Q_{desc}</i>	Root	19,456,928.3	147.7	140.6	130Kx	140Kx	261.2	91.7	77.4	2.8x	3.4x	4,319.7	5,255.7	5,310.4	0.82x	0.81x
NT100K	<i>Q_{desc}</i>	Leaf parent H	2,542.6	138.0	124.3	18x	20x	3.8	50.8	22.6	0.07x	0.17x	2.1	2.5	3.3	0.86x	0.65x
NT100K	<i>Q_{desc}</i>	Leaf parent	469.9	137.8	114.2	3.4x	4.1x	2.5	51.4	26.9	0.05x	0.09x	1.4	1.6	1.9	0.90x	0.77x
DT10	<i>Q_{desc}</i>	Root	9.5	1.4	1.4	7.0x	6.7x	2.9	1.5	1.7	2.0x	1.7x	1.2	2.1	2.2	0.54x	0.53x
DT10	<i>Q_{desc}</i>	Leaf parent H	9.3	1.4	1.4	6.8x	6.8x	2.8	1.5	1.6	1.8x	1.8x	1.7	1.9	1.9	0.91x	0.89x
DT10	<i>Q_{desc}</i>	Leaf parent	9.0	1.4	1.4	6.6x	6.7x	2.8	1.6	1.6	1.8x	1.7x	1.3	1.3	1.4	0.97x	0.97x
DT100	<i>Q_{desc}</i>	Root	20.4	1.5	1.5	14x	14x	3.9	1.5	1.8	2.6x	2.2x	11.6	13.4	13.8	0.86x	0.84x
DT100	<i>Q_{desc}</i>	Leaf parent H	9.3	1.4	1.4	6.5x	6.6x	3.0	1.5	1.7	2.0x	1.8x	1.7	1.8	1.8	0.95x	0.95x
DT100	<i>Q_{desc}</i>	Leaf parent	9.3	1.4	1.5	6.5x	6.4x	2.8	1.6	1.6	1.8x	1.7x	1.5	1.6	1.6	0.94x	0.92x
DT1K	<i>Q_{desc}</i>	Root	1,107.9	2.8	2.8	390x	390x	11.1	1.7	1.8	6.6x	6.2x	39.8	47.3	49.0	0.84x	0.81x
DT1K	<i>Q_{desc}</i>	Leaf parent H	13.7	2.3	2.2	5.9x	6.2x	3.5	1.7	1.6	2.1x	2.2x	0.9	0.9	0.9	0.95x	0.99x
DT1K	<i>Q_{desc}</i>	Leaf parent	10.9	2.3	2.2	4.7x	4.9x	3.7	1.6	1.5	2.4x	2.4x	0.7	0.8	1.0	0.92x	0.69x
DT10K	<i>Q_{desc}</i>	Root	107,336.6	15.9	15.4	6.7Kx	7.0Kx	32.1	7.1	8.8	4.5x	3.6x	417.6	493.1	505.9	0.85x	0.82x
DT10K	<i>Q_{desc}</i>	Leaf parent H	246.8	11.6	9.8	21x	25x	2.6	4.6	2.7	0.56x	0.98x	1.2	1.4	1.6	0.89x	0.77x
DT10K	<i>Q_{desc}</i>	Leaf parent	223.2	11.5	9.2	19x	24x	2.6	5.3	2.8	0.49x	0.92x	1.4	1.5	1.2	0.93x	1.2x
DT100K	<i>Q_{desc}</i>	Root	>6 h	143.3	141.4	>150Kx	>150Kx	134.7	92.6	77.2	1.5x	1.7x	4,296.9	5,152.6	5,355.2	0.83x	0.80x
DT100K	<i>Q_{desc}</i>	Leaf parent H	801.4	143.8	120.8	5.6x	6.6x	3.4	51.4	26.6	0.07x	0.13x	1.3	1.8	1.8	0.74x	0.76x
DT100K	<i>Q_{desc}</i>	Leaf parent	473.4	138.3	120.3	3.4x	3.9x	2.3	51.0	13.4	0.04x	0.17x	1.6	1.6	1.6	0.99x	1.0x

Queries: *Q_{desc}* = All Descendants; *Q_{child}* = All Children; *Q_{leaf}* = All Leaves; *Q_{a&d}* = Check if Ancestor.

Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.

Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.

Timeouts: each >T runtime cell gives the user-confirmed limit T for that row and method (unit shown).

Speedup bounds use those limits in the same row/system; > is a lower bound, < an upper bound.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Descendants (Query 01)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
WT10	<i>Q_{desc}</i>	Root	9.4	1.4	1.4	6.8x	6.6x	3.0	1.6	1.5	1.9x	2.0x	2.0	2.2	2.2	0.93x	0.91x
WT10	<i>Q_{desc}</i>	Leaf parent H	9.3	1.4	1.3	6.6x	7.0x	3.1	1.6	1.5	1.9x	2.1x	1.7	1.9	2.0	0.89x	0.85x
WT10	<i>Q_{desc}</i>	Leaf parent	9.0	1.4	1.4	6.6x	6.6x	2.8	1.4	1.5	2.0x	1.9x	1.4	1.5	1.5	0.97x	0.96x
WT100	<i>Q_{desc}</i>	Root	26.0	2.2	1.8	12x	14x	3.5	1.5	1.5	2.3x	2.3x	11.6	13.2	13.5	0.88x	0.86x
WT100	<i>Q_{desc}</i>	Leaf parent H	10.3	1.4	1.4	7.1x	7.1x	2.7	1.7	1.4	1.6x	1.9x	2.5	2.7	2.8	0.93x	0.91x
WT100	<i>Q_{desc}</i>	Leaf parent	11.7	1.7	1.7	6.7x	6.8x	3.0	1.5	1.5	2.0x	2.0x	1.7	2.1	1.8	0.82x	0.95x
WT1K	<i>Q_{desc}</i>	Root	1,152.5	2.6	2.7	440x	420x	8.7	1.6	2.2	5.3x	3.9x	58.8	45.1	46.8	1.3x	1.3x
WT1K	<i>Q_{desc}</i>	Leaf parent H	20.2	2.3	2.3	8.8x	8.7x	3.3	1.9	1.8	1.7x	1.8x	1.6	2.0	2.6	0.77x	0.60x
WT1K	<i>Q_{desc}</i>	Leaf parent	21.9	2.3	1.7	9.7x	13x	3.5	1.6	1.7	2.2x	2.0x	2.2	2.3	2.5	0.93x	0.88x
WT10K	<i>Q_{desc}</i>	Root	112,250.6	8.6	10.9	13Kx	10Kx	24.9	10.3	8.6	2.4x	2.9x	407.5	486.5	503.7	0.84x	0.81x
WT10K	<i>Q_{desc}</i>	Leaf parent H	332.9	11.6	10.1	29x	33x	2.4	5.2	3.0	0.47x	0.81x	1.7	2.2	2.8	0.78x	0.61x
WT10K	<i>Q_{desc}</i>	Leaf parent	114.7	5.7	1.9	20x	60x	3.0	4.6	2.9	0.65x	1.0x	2.4	2.6	2.6	0.92x	0.93x
WT100K	<i>Q_{desc}</i>	Root	23,750,824.0	76.4	74.2	310Kx	320Kx	122.5	91.6	75.6	1.3x	1.6x	4,313.0	5,124.1	5,269.5	0.84x	0.82x
WT100K	<i>Q_{desc}</i>	Leaf parent H	2,976.6	139.2	126.7	21x	23x	3.5	50.0	26.5	0.07x	0.13x	2.3	2.6	2.6	0.89x	0.90x
WT100K	<i>Q_{desc}</i>	Leaf parent	575.1	37.4	1.9	15x	300x	2.2	50.7	26.9	0.04x	0.08x	1.2	1.3	1.3	0.93x	0.92x
SNB/C	<i>Q_{desc}</i>	Root	91,997.5	681.0	2.0	140x	47Kx	3.6	417.0	391.4	0.01x	0.01x	4.7	4.5	4.7	1.0x	1.0x
SNB/C	<i>Q_{desc}</i>	Deep root	81,661.1	2,631.1	2,120.5	31x	39x	4.1	393.3	400.4	0.01x	0.01x	4.1	4.4	4.1	0.92x	1.0x
SNB/C	<i>Q_{desc}</i>	Leaf parent H	68,727.3	2,615.8	2,430.7	26x	28x	3.6	410.0	366.1	0.01x	0.01x	3.5	3.8	4.0	0.93x	0.87x
SNB/C	<i>Q_{desc}</i>	Leaf parent	22,391.0	724.1	2.0	31x	11Kx	3.4	436.1	392.9	0.01x	0.01x	1.6	1.6	1.6	1.0x	0.99x
SNB/P	<i>Q_{desc}</i>	Root	16,838.8	2.7	2.6	6.2Kx	6.6Kx	6.9	2.5	2.3	2.8x	3.1x	46.3	41.2	42.9	1.1x	1.1x
SNB/P	<i>Q_{desc}</i>	Leaf parent H	15,130.5	3.0	2.9	5.1Kx	5.1Kx	3.7	2.0	1.8	1.9x	2.0x	10.9	12.8	13.8	0.85x	0.79x
SNB/P	<i>Q_{desc}</i>	Leaf parent	14,506.6	1.6	1.9	9.0Kx	7.8Kx	3.8	2.1	1.9	1.8x	2.0x	1.1	1.3	1.5	0.87x	0.74x
SNB/T	<i>Q_{desc}</i>	Root	14,465.3	1.9	1.9	7.7Kx	7.6Kx	3.3	1.7	1.6	2.0x	2.1x	9.7	11.0	11.3	0.89x	0.85x
SNB/T	<i>Q_{desc}</i>	Leaf parent H	14,542.6	1.5	1.5	9.9Kx	10Kx	3.1	1.6	1.7	1.9x	1.8x	8.6	8.9	9.1	0.96x	0.94x
SNB/T	<i>Q_{desc}</i>	Leaf parent	14,497.3	1.6	1.7	9.0Kx	8.6Kx	2.9	1.6	1.5	1.9x	2.0x	1.5	1.5	1.5	1.0x	1.0x

Queries: *Q_{desc}* = All Descendants; *Q_{child}* = All Children; *Q_{leaf}* = All Leaves; *Q_{a&d}* = Check if Ancestor.

Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.

Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Children (Query 02)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
F40	<i>Q_{child}</i>	High degree	2.0	1.6	1.5	1.2x	1.3x	1.7	1.8	1.8	0.91x	0.93x	1.7	1.6	1.6	1.1x	1.0x
F40	<i>Q_{child}</i>	Low degree	1.9	1.6	1.5	1.2x	1.3x	2.0	1.7	1.9	1.2x	1.0x	1.2	1.1	1.1	1.1x	1.1x
F1K	<i>Q_{child}</i>	High degree	2.4	2.5	2.3	0.97x	1.0x	1.6	2.5	2.5	0.63x	0.64x	1.9	1.9	1.6	0.99x	1.2x
F1K	<i>Q_{child}</i>	Low degree	2.3	2.4	2.2	0.95x	1.0x	1.5	2.3	1.4	0.65x	1.1x	1.1	1.1	1.1	0.97x	0.96x
NT10	<i>Q_{child}</i>	High degree	1.9	1.6	1.3	1.2x	1.5x	1.7	2.0	1.9	0.82x	0.89x	1.6	1.7	1.7	0.94x	0.94x
NT10	<i>Q_{child}</i>	Low degree	1.9	1.6	1.4	1.2x	1.3x	1.6	1.8	1.3	0.94x	1.3x	1.3	1.1	1.1	1.2x	1.2x
NT100	<i>Q_{child}</i>	High degree	2.0	1.5	1.5	1.3x	1.3x	1.8	1.7	1.7	1.1x	1.1x	2.0	2.1	2.2	0.97x	0.91x
NT100	<i>Q_{child}</i>	Low degree	1.9	1.6	1.5	1.2x	1.3x	1.7	1.7	1.8	1.0x	0.95x	1.3	1.2	1.4	1.1x	0.89x
NT1K	<i>Q_{child}</i>	High degree	2.4	2.4	2.3	1.0x	1.0x	1.7	1.9	2.0	0.89x	0.88x	1.2	1.1	1.1	1.1x	1.1x
NT1K	<i>Q_{child}</i>	Low degree	2.2	2.3	2.2	0.97x	1.0x	1.7	1.9	1.8	0.91x	0.96x	0.7	0.7	0.7	0.97x	1.0x
NT10K	<i>Q_{child}</i>	High degree	6.3	10.5	10.4	0.60x	0.61x	2.2	4.4	3.0	0.49x	0.72x	1.9	1.9	1.9	0.98x	1.0x
NT10K	<i>Q_{child}</i>	Low degree	5.4	10.4	10.3	0.52x	0.53x	2.2	5.1	2.4	0.43x	0.91x	1.1	1.2	1.1	0.94x	0.97x
NT100K	<i>Q_{child}</i>	High degree	57.3	107.3	104.0	0.53x	0.55x	2.2	38.9	27.0	0.06x	0.08x	3.5	3.9	4.1	0.92x	0.86x
NT100K	<i>Q_{child}</i>	Low degree	55.8	105.0	103.7	0.53x	0.54x	1.9	38.6	26.7	0.05x	0.07x	1.3	1.4	1.3	0.93x	0.98x
DT10	<i>Q_{child}</i>	High degree	2.0	1.6	1.5	1.3x	1.3x	1.8	1.4	1.5	1.3x	1.2x	1.5	1.7	1.6	0.89x	0.95x
DT10	<i>Q_{child}</i>	Low degree	1.9	1.6	1.5	1.2x	1.3x	1.6	1.8	1.7	0.90x	0.92x	1.1	1.1	1.1	0.97x	0.97x
DT100	<i>Q_{child}</i>	High degree	2.0	1.6	1.6	1.2x	1.3x	1.4	1.4	1.7	0.99x	0.82x	1.7	1.6	1.5	1.1x	1.1x
DT100	<i>Q_{child}</i>	Low degree	1.9	1.6	1.5	1.2x	1.3x	1.9	1.6	1.8	1.2x	1.1x	1.0	1.1	1.1	0.94x	0.94x
DT1K	<i>Q_{child}</i>	High degree	2.4	2.4	2.3	1.0x	1.1x	1.1	1.8	1.9	0.64x	0.60x	1.5	1.5	1.4	0.98x	1.1x
DT1K	<i>Q_{child}</i>	Low degree	2.3	2.4	2.3	0.94x	1.0x	1.8	1.8	1.8	1.0x	1.0x	1.0	1.0	1.0	0.93x	0.92x
DT10K	<i>Q_{child}</i>	High degree	6.3	10.5	10.3	0.60x	0.61x	1.7	4.9	2.8	0.34x	0.59x	1.5	1.5	1.5	1.0x	1.0x
DT10K	<i>Q_{child}</i>	Low degree	5.4	10.4	10.2	0.52x	0.53x	1.8	4.3	2.6	0.41x	0.67x	1.0	1.0	1.0	1.0x	1.0x
DT100K	<i>Q_{child}</i>	High degree	65.5	105.0	99.8	0.62x	0.66x	1.7	39.2	20.3	0.04x	0.08x	1.8	1.7	1.7	1.0x	1.1x
DT100K	<i>Q_{child}</i>	Low degree	56.3	107.1	103.9	0.53x	0.54x	1.8	38.9	20.2	0.05x	0.09x	1.2	1.3	1.2	0.90x	1.0x

Queries: *Q_{desc}* = All Descendants; *Q_{child}* = All Children; *Q_{leaf}* = All Leaves; *Q_{a&d}* = Check if Ancestor.

Parameters: High degree = highest-degree node; Low degree = lowest-degree node.

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Children (Query 02)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
WT10	<i>Q_{child}</i>	High degree	2.0	1.6	1.5	1.3x	1.4x	2.2	1.8	1.7	1.2x	1.3x	1.7	1.7	1.7	0.97x	0.99x
WT10	<i>Q_{child}</i>	Low degree	1.8	1.6	1.4	1.2x	1.3x	1.7	1.6	1.3	1.0x	1.3x	1.1	1.1	1.2	0.99x	0.95x
WT100	<i>Q_{child}</i>	High degree	2.0	1.7	1.6	1.2x	1.3x	1.6	1.7	1.8	0.92x	0.87x	2.0	2.1	1.8	0.91x	1.1x
WT100	<i>Q_{child}</i>	Low degree	2.0	1.7	1.5	1.2x	1.3x	1.6	1.4	1.5	1.1x	1.1x	1.1	1.2	1.2	0.98x	0.96x
WT1K	<i>Q_{child}</i>	High degree	2.5	2.4	2.4	1.0x	1.0x	1.6	1.8	1.8	0.89x	0.86x	2.3	2.5	2.4	0.95x	0.99x
WT1K	<i>Q_{child}</i>	Low degree	2.3	2.3	2.3	1.0x	0.99x	1.6	1.7	1.6	0.93x	1.0x	1.1	1.2	1.2	0.95x	0.92x
WT10K	<i>Q_{child}</i>	High degree	6.3	10.6	10.4	0.59x	0.60x	1.5	5.0	2.9	0.29x	0.51x	2.6	2.9	2.7	0.88x	0.96x
WT10K	<i>Q_{child}</i>	Low degree	5.4	10.3	10.3	0.52x	0.52x	1.5	4.2	2.6	0.35x	0.58x	1.2	1.2	1.2	0.98x	0.98x
WT100K	<i>Q_{child}</i>	High degree	63.0	101.7	99.6	0.62x	0.63x	1.6	38.9	26.6	0.04x	0.06x	1.5	1.8	1.8	0.87x	0.84x
WT100K	<i>Q_{child}</i>	Low degree	54.2	104.9	99.3	0.52x	0.55x	2.0	38.6	22.4	0.05x	0.09x	0.7	0.7	0.7	1.1x	1.1x
SNB/C	<i>Q_{child}</i>	High degree	1,427.9	3,579.4	3,494.6	0.40x	0.41x	2.8	372.8	387.5	0.01x	0.01x	3.4	3.8	3.8	0.90x	0.89x
SNB/C	<i>Q_{child}</i>	Low degree	1,169.3	3,542.6	3,203.2	0.33x	0.36x	1.8	362.5	356.7	0.01x	0.01x	1.2	1.3	1.0	0.92x	1.2x
SNB/P	<i>Q_{child}</i>	High degree	3.5	3.3	3.1	1.1x	1.2x	2.4	2.3	2.1	1.0x	1.1x	24.4	25.6	26.3	0.95x	0.93x
SNB/P	<i>Q_{child}</i>	Low degree	2.4	2.9	2.5	0.82x	0.94x	1.5	2.4	2.4	0.62x	0.63x	1.2	1.3	1.3	0.91x	0.92x
SNB/T	<i>Q_{child}</i>	High degree	2.1	1.7	1.6	1.3x	1.4x	1.7	1.6	1.8	1.0x	0.94x	3.7	4.2	4.1	0.89x	0.90x
SNB/T	<i>Q_{child}</i>	Low degree	1.9	1.6	1.5	1.2x	1.3x	1.7	1.5	1.7	1.1x	1.0x	1.2	1.1	1.3	1.1x	0.94x

Queries: *Q_{desc}* = All Descendants; *Q_{child}* = All Children; *Q_{leaf}* = All Leaves; *Q_{a&d}* = Check if Ancestor.

Parameters: High degree = highest-degree node; Low degree = lowest-degree node.

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Leaves (Query 05)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
F40	<i>Qleaf</i>	Root	9.6	1.6	2.1	6.1x	4.7x	5.3	3.1	1.7	1.7x	3.2x	1.9	1.9	1.7	1.0x	1.1x
F40	<i>Qleaf</i>	Leaf parent H	9.9	1.5	2.0	6.5x	4.8x	4.6	2.7	2.0	1.7x	2.3x	1.4	1.5	1.5	0.94x	0.94x
F40	<i>Qleaf</i>	Leaf parent	9.6	1.5	2.0	6.5x	4.7x	4.2	3.2	1.9	1.3x	2.2x	1.4	1.5	1.7	0.96x	0.84x
F1K	<i>Qleaf</i>	Root	305.9	19.0	2.4	16x	130x	8.0	3.0	2.2	2.7x	3.6x	11.3	26.4	26.3	0.43x	0.43x
F1K	<i>Qleaf</i>	Leaf parent H	96.2	2.4	2.3	40x	43x	5.6	3.5	1.9	1.6x	2.9x	2.3	2.7	2.1	0.86x	1.1x
F1K	<i>Qleaf</i>	Leaf parent	95.0	2.4	2.3	40x	42x	6.4	2.6	1.8	2.4x	3.6x	1.9	1.9	1.7	0.96x	1.1x
NT10	<i>Qleaf</i>	Root	9.7	1.5	1.4	6.6x	6.9x	3.8	2.8	1.9	1.4x	2.0x	1.8	1.9	1.8	0.96x	0.98x
NT10	<i>Qleaf</i>	Leaf parent H/L	9.0	1.4	1.4	6.3x	6.3x	4.6	2.9	1.9	1.6x	2.4x	1.6	1.7	1.5	0.93x	1.1x
NT100	<i>Qleaf</i>	Root	18.0	2.8	6.3	6.4x	2.9x	6.1	3.1	1.6	2.0x	3.8x	9.4	11.3	9.8	0.84x	0.96x
NT100	<i>Qleaf</i>	Leaf parent H	11.2	1.5	6.1	7.4x	1.8x	5.0	2.9	1.8	1.7x	2.8x	2.2	2.4	2.1	0.91x	1.1x
NT100	<i>Qleaf</i>	Leaf parent	10.9	1.5	6.1	7.2x	1.8x	5.4	2.9	1.6	1.8x	3.3x	1.8	1.8	1.6	1.0x	1.1x
NT1K	<i>Qleaf</i>	Root	946.0	126.1	2.8	7.5x	340x	12.5	3.2	2.2	3.9x	5.6x	30.6	34.6	35.1	0.88x	0.87x
NT1K	<i>Qleaf</i>	Leaf parent H	161.4	2.3	2.3	71x	70x	4.5	3.6	2.0	1.2x	2.2x	1.4	1.7	1.5	0.86x	0.96x
NT1K	<i>Qleaf</i>	Leaf parent	158.7	2.5	2.3	65x	69x	5.2	3.3	1.7	1.5x	3.0x	1.2	1.2	1.1	1.0x	1.1x
NT10K	<i>Qleaf</i>	Root	92,769.6	12,618.4	14.5	7.4x	6.4Kx	48.5	20.5	7.6	2.4x	6.4x	294.7	338.5	349.4	0.87x	0.84x
NT10K	<i>Qleaf</i>	Leaf parent H	11,983.1	11.6	10.8	1.0Kx	1.1Kx	5.7	17.6	2.4	0.32x	2.3x	1.4	1.9	2.1	0.75x	0.66x
NT10K	<i>Qleaf</i>	Leaf parent	11,977.1	11.7	10.8	1.0Kx	1.1Kx	5.1	16.2	2.3	0.32x	2.2x	1.8	1.9	1.9	0.92x	0.96x
NT100K	<i>Qleaf</i>	Root	19,008,309.3	1,153,555.7	113.8	16x	170Kx	108.7	203.3	74.1	0.53x	1.5x	4,085.6	4,919.9	5,057.6	0.83x	0.81x
NT100K	<i>Qleaf</i>	Leaf parent H	1,230,113.2	339.5	112.0	3.6Kx	11Kx	5.7	156.1	21.1	0.04x	0.27x	2.4	2.5	2.4	0.97x	1.0x
NT100K	<i>Qleaf</i>	Leaf parent	1,227,753.7	338.8	112.3	3.6Kx	11Kx	4.6	167.3	20.2	0.03x	0.23x	1.3	1.2	1.0	1.0x	1.2x
DT10	<i>Qleaf</i>	Root	9.4	1.5	1.4	6.2x	6.5x	3.9	2.7	1.7	1.5x	2.3x	1.8	1.8	2.0	0.97x	0.90x
DT10	<i>Qleaf</i>	Leaf parent H	9.6	1.4	1.4	6.8x	6.7x	3.6	2.6	1.6	1.4x	2.2x	2.0	1.9	1.9	1.0x	1.0x
DT10	<i>Qleaf</i>	Leaf parent	9.7	1.3	1.5	7.4x	6.6x	3.9	2.6	1.7	1.5x	2.3x	1.5	1.6	1.5	0.95x	1.0x
DT100	<i>Qleaf</i>	Root	16.2	2.8	5.4	5.7x	3.0x	4.5	2.8	1.8	1.6x	2.5x	6.8	9.0	8.2	0.75x	0.83x
DT100	<i>Qleaf</i>	Leaf parent H	11.1	1.5	5.2	7.3x	2.1x	4.1	2.5	1.7	1.6x	2.5x	1.8	1.9	1.8	0.94x	1.0x
DT100	<i>Qleaf</i>	Leaf parent	10.8	1.5	5.3	7.1x	2.0x	4.1	2.9	1.8	1.4x	2.3x	1.6	1.6	1.6	0.96x	1.0x
DT1K	<i>Qleaf</i>	Root	645.4	119.8	2.6	5.4x	240x	13.1	2.8	2.0	4.7x	6.5x	37.6	33.8	30.0	1.1x	1.3x
DT1K	<i>Qleaf</i>	Leaf parent H	97.0	2.4	2.3	40x	43x	4.8	3.2	2.0	1.5x	2.4x	1.3	1.9	1.9	0.72x	0.72x
DT1K	<i>Qleaf</i>	Leaf parent	96.3	2.4	2.3	41x	42x	5.1	2.9	1.8	1.8x	2.8x	1.6	1.7	1.6	0.99x	1.0x
DT10K	<i>Qleaf</i>	Root	59,616.8	10,885.1	13.1	5.5x	4.5Kx	30.4	10.6	5.3	2.9x	5.7x	197.5	226.7	229.5	0.87x	0.86x
DT10K	<i>Qleaf</i>	Leaf parent H	6,430.0	11.7	10.5	550x	610x	3.8	13.2	1.8	0.29x	2.1x	1.3	1.6	1.8	0.81x	0.70x
DT10K	<i>Qleaf</i>	Leaf parent	6,417.6	11.7	10.5	550x	610x	5.2	17.5	1.9	0.29x	2.7x	1.6	1.6	1.6	1.0x	1.0x
DT100K	<i>Qleaf</i>	Root	11,677,527.2	896,628.6	97.4	13x	120Kx	135.9	156.5	42.1	0.87x	3.2x	2,118.7	2,564.3	2,614.5	0.83x	0.81x
DT100K	<i>Qleaf</i>	Leaf parent H	617,168.0	344.4	110.8	1.8Kx	5.6Kx	5.9	130.1	11.6	0.05x	0.51x	1.3	1.4	1.3	0.88x	0.98x
DT100K	<i>Qleaf</i>	Leaf parent	617,092.0	341.9	108.5	1.8Kx	5.7Kx	4.5	120.9	12.4	0.04x	0.36x	1.1	1.2	1.0	0.94x	1.1x

Queries: *Qdesc* = All Descendants; *Qchild* = All Children; *Qleaf* = All Leaves; *Qa&d* = Check if Ancestor.

Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.

Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Leaves (Query 05)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
WT10	<i>Qleaf</i>	Root	9.6	1.5	1.5	6.4x	6.5x	4.1	2.6	1.6	1.5x	2.6x	1.9	1.9	1.9	0.96x	1.0x
WT10	<i>Qleaf</i>	Leaf parent H	9.2	1.4	1.4	6.5x	6.4x	4.3	2.9	1.9	1.5x	2.3x	1.6	1.9	1.8	0.88x	0.90x
WT10	<i>Qleaf</i>	Leaf parent	9.6	1.4	1.4	6.7x	6.7x	3.7	2.5	1.5	1.5x	2.5x	1.5	1.6	1.5	0.94x	0.99x
WT100	<i>Qleaf</i>	Root	20.3	2.9	7.6	7.1x	2.7x	5.0	3.0	1.9	1.7x	2.7x	7.3	11.3	12.0	0.65x	0.61x
WT100	<i>Qleaf</i>	Leaf parent H	12.0	1.6	7.5	7.6x	1.6x	3.7	2.7	1.9	1.4x	1.9x	2.8	3.0	3.0	0.91x	0.94x
WT100	<i>Qleaf</i>	Leaf parent	11.1	1.6	7.3	7.1x	1.5x	3.8	2.7	1.9	1.4x	2.0x	2.1	2.2	2.1	0.94x	0.98x
WT1K	<i>Qleaf</i>	Root	1,134.1	134.9	2.8	8.4x	410x	9.4	3.2	1.9	3.0x	4.9x	45.1	40.3	41.9	1.1x	1.1x
WT1K	<i>Qleaf</i>	Leaf parent H	169.6	2.4	2.3	70x	73x	4.7	3.5	1.9	1.4x	2.4x	1.7	1.8	1.7	0.94x	0.98x
WT1K	<i>Qleaf</i>	Leaf parent	167.4	2.4	2.3	69x	72x	4.8	3.4	1.6	1.4x	3.0x	1.5	1.6	1.6	0.94x	0.98x
WT10K	<i>Qleaf</i>	Root	110,118.3	13,540.2	15.3	8.1x	7.2Kx	29.8	25.3	8.1	1.2x	3.7x	376.1	400.6	461.0	0.94x	0.82x
WT10K	<i>Qleaf</i>	Leaf parent H	12,540.9	11.8	10.9	1.1Kx	1.1Kx	3.7	21.1	2.5	0.18x	1.5x	2.1	2.6	3.2	0.78x	0.64x
WT10K	<i>Qleaf</i>	Leaf parent	12,518.4	11.6	10.9	1.1Kx	1.1Kx	4.5	21.0	2.5	0.21x	1.8x	2.8	2.7	2.2	1.0x	1.3x
WT100K	<i>Qleaf</i>	Root	18,195,767.1	1,084,120.5	114.5	17x	160Kx	114.9	197.6	73.5	0.58x	1.6x	3,990.9	4,720.8	4,920.8	0.84x	0.81x
WT100K	<i>Qleaf</i>	Leaf parent H	1,227,203.3	341.3	105.6	3.6Kx	12Kx	5.4	160.9	26.4	0.03x	0.21x	2.3	3.0	4.3	0.78x	0.54x
WT100K	<i>Qleaf</i>	Leaf parent	1,224,686.1	337.4	111.8	3.6Kx	11Kx	4.3	171.8	20.7	0.03x	0.21x	1.8	2.0	1.9	0.90x	0.94x
SNB/C	<i>Qleaf</i>	Root	>6 h	2,686.8	2,345.2	>8.0Kx	>9.2Kx	6.0	552.9	326.0	0.01x	0.02x	2.0	3.1	2.9	0.66x	0.70x
SNB/C	<i>Qleaf</i>	Deep root	>6 h	2,719.6	2,333.1	>7.9Kx	>9.2Kx	5.9	569.7	293.9	0.01x	0.02x	2.7	2.9	3.1	0.91x	0.86x
SNB/C	<i>Qleaf</i>	Leaf parent H	>6 h	2,685.2	2,340.8	>8.0Kx	>9.2Kx	5.2	592.0	339.9	0.01x	0.01x	3.6	3.9	3.9	0.93x	0.93x
SNB/C	<i>Qleaf</i>	Leaf parent	>6 h	2,718.6	2,332.8	>7.9Kx	>9.2Kx	5.9	584.0	342.6	0.01x	0.02x	1.6	1.7	1.6	0.98x	1.0x
SNB/P	<i>Qleaf</i>	Root	16,989.0	80.9	3.2	210x	5.3Kx	9.6	4.3	2.5	2.2x	3.8x	46.1	40.2	41.5	1.1x	1.1x
SNB/P	<i>Qleaf</i>	Leaf parent H	15,466.6	8.8	3.0	1.8Kx	5.2Kx	5.1	4.0	2.1	1.3x	2.4x	11.7	14.3	15.4	0.81x	0.76x
SNB/P	<i>Qleaf</i>	Leaf parent	14,814.6	2.9	2.8	5.1Kx	5.3Kx	5.0	4.2	2.3	1.2x	2.2x	1.2	1.5	1.6	0.80x	0.80x
SNB/T	<i>Qleaf</i>	Root	14,533.6	2.3	4.4	6.4Kx	3.3Kx	4.6	2.5	1.8	1.9x	2.6x	10.2	9.8	9.5	1.0x	1.1x
SNB/T	<i>Qleaf</i>	Leaf parent H	14,546.1	2.0	4.3	7.3Kx	3.4Kx	4.4	2.6	1.7	1.7x	2.6x	8.2	6.1	8.6	1.3x	0.95x
SNB/T	<i>Qleaf</i>	Leaf parent	14,430.3	1.5	4.2	9.5Kx	3.4Kx	3.4	2.4	1.7	1.4x	2.0x	1.7	1.5	1.5	1.1x	1.1x

Queries: *Qdesc* = All Descendants; *Qchild* = All Children; *Qleaf* = All Leaves; *Qand* = Check if Ancestor.

Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.

Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.

Timeouts: each >T runtime cell gives the user-confirmed limit T for that row and method (unit shown).

Speedup bounds use those limits in the same row/system; > is a lower bound, < an upper bound.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

Check if Ancestor (Query 11)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
F40	<i>Qa&d</i>	Root-leaf	10.1	1.4	1.3	7.3x	7.9x	22.2	0.9	0.9	25x	24x	1.5	1.5	1.3	1.0x	1.1x
F40	<i>Qa&d</i>	Deep parent-leaf	10.1	1.1	1.2	9.2x	8.1x	19.9	0.9	0.9	22x	21x	1.3	1.3	1.5	0.98x	0.87x
F40	<i>Qa&d</i>	Siblings	9.9	1.1	1.1	9.0x	9.1x	20.1	1.0	0.9	21x	22x	1.5	1.3	1.3	1.2x	1.1x
F40	<i>Qa&d</i>	Distant leaves	9.3	1.4	1.1	6.7x	8.5x	21.4	0.9	0.9	24x	23x	1.5	1.4	1.4	1.1x	1.0x
F1K	<i>Qa&d</i>	Root-leaf	12.5	1.4	1.5	8.8x	8.4x	19.1	1.0	0.9	19x	21x	1.7	1.1	1.6	1.5x	1.1x
F1K	<i>Qa&d</i>	Deep parent-leaf	12.2	1.6	1.5	7.5x	8.4x	21.6	1.0	0.9	22x	24x	1.8	1.2	1.6	1.6x	1.2x
F1K	<i>Qa&d</i>	Siblings	10.8	2.6	2.6	4.2x	4.2x	16.9	1.1	0.9	16x	19x	1.6	1.3	1.6	1.2x	0.96x
F1K	<i>Qa&d</i>	Distant leaves	10.4	2.5	2.6	4.1x	4.0x	22.3	1.0	1.0	23x	21x	1.7	1.3	1.6	1.3x	1.0x
NT10	<i>Qa&d</i>	Root-leaf	9.8	1.2	1.1	8.0x	9.1x	18.7	0.9	0.9	20x	20x	1.6	1.3	1.2	1.2x	1.3x
NT10	<i>Qa&d</i>	Deep parent-leaf	9.8	1.1	1.2	8.9x	7.9x	19.0	1.0	1.0	19x	20x	1.3	1.2	1.1	1.1x	1.2x
NT10	<i>Qa&d</i>	Siblings	9.3	1.3	1.0	7.3x	9.5x	20.9	1.0	1.0	21x	21x	1.5	1.4	1.7	1.0x	0.87x
NT10	<i>Qa&d</i>	Distant leaves	9.4	1.0	1.0	9.4x	9.4x	20.4	0.9	0.9	23x	22x	1.4	1.2	1.2	1.2x	1.1x
NT100	<i>Qa&d</i>	Root-leaf	10.0	1.3	1.3	7.6x	7.6x	20.7	0.9	0.9	22x	23x	2.0	1.4	0.8	1.5x	2.7x
NT100	<i>Qa&d</i>	Deep parent-leaf	10.1	1.3	1.3	7.9x	7.7x	20.7	0.9	1.0	22x	21x	1.4	1.3	1.5	1.1x	0.96x
NT100	<i>Qa&d</i>	Siblings	9.3	1.2	1.1	8.0x	8.2x	18.7	0.9	0.9	20x	20x	1.7	1.3	1.2	1.3x	1.5x
NT100	<i>Qa&d</i>	Distant leaves	9.5	1.2	1.1	8.3x	8.5x	21.8	0.9	0.9	24x	23x	1.2	0.9	0.9	1.3x	1.3x
NT1K	<i>Qa&d</i>	Root-leaf	10.8	1.5	1.4	7.4x	7.9x	19.5	1.0	0.9	20x	22x	1.2	0.8	0.7	1.5x	1.7x
NT1K	<i>Qa&d</i>	Deep parent-leaf	10.6	1.4	1.3	7.6x	8.1x	19.3	0.8	0.9	23x	22x	0.9	1.1	1.1	0.78x	0.79x
NT1K	<i>Qa&d</i>	Siblings	10.8	2.6	2.8	4.2x	3.9x	22.4	1.0	0.9	24x	26x	1.4	1.1	1.2	1.3x	1.2x
NT1K	<i>Qa&d</i>	Distant leaves	10.6	2.5	2.6	4.2x	4.1x	20.6	0.9	0.9	23x	22x	1.4	1.1	0.9	1.4x	1.5x
NT10K	<i>Qa&d</i>	Root-leaf	18.2	1.6	1.4	12x	13x	22.5	0.9	0.9	26x	24x	1.6	1.3	1.4	1.2x	1.2x
NT10K	<i>Qa&d</i>	Deep parent-leaf	18.1	1.5	1.6	12x	11x	22.5	0.9	0.9	26x	26x	1.3	1.1	1.2	1.2x	1.2x
NT10K	<i>Qa&d</i>	Siblings	21.2	16.4	16.3	1.3x	1.3x	21.0	0.9	0.9	24x	23x	1.4	1.2	1.3	1.1x	1.1x
NT10K	<i>Qa&d</i>	Distant leaves	21.0	16.5	16.3	1.3x	1.3x	24.7	0.8	0.9	30x	26x	1.6	1.2	1.2	1.3x	1.3x
NT100K	<i>Qa&d</i>	Root-leaf	78.5	1.6	1.6	49x	48x	19.8	1.1	0.9	18x	23x	1.1	1.0	0.8	1.2x	1.4x
NT100K	<i>Qa&d</i>	Deep parent-leaf	78.7	1.6	1.6	49x	49x	22.2	0.9	0.9	23x	24x	0.9	0.8	0.8	1.2x	1.1x
NT100K	<i>Qa&d</i>	Siblings	125.4	157.2	157.8	0.80x	0.80x	18.4	1.0	0.9	18x	21x	0.8	0.8	0.8	1.0x	1.0x
NT100K	<i>Qa&d</i>	Distant leaves	125.5	159.4	158.8	0.79x	0.79x	19.7	1.0	0.9	19x	21x	0.9	0.8	0.8	1.1x	1.1x
DT10	<i>Qa&d</i>	Root-leaf	9.8	1.3	1.3	7.3x	7.4x	17.4	0.9	1.0	20x	18x	1.4	1.2	1.1	1.2x	1.3x
DT10	<i>Qa&d</i>	Deep parent-leaf	9.7	1.1	1.3	9.0x	7.6x	18.4	0.9	1.0	21x	18x	1.3	1.2	1.2	1.1x	1.1x
DT10	<i>Qa&d</i>	Siblings	9.4	1.1	1.0	8.8x	9.2x	19.1	0.9	1.0	22x	20x	1.2	1.2	1.2	1.0x	1.0x
DT10	<i>Qa&d</i>	Distant leaves	9.4	1.0	1.0	9.2x	9.6x	18.0	0.9	0.9	21x	20x	1.3	1.2	1.2	1.1x	1.1x
DT100	<i>Qa&d</i>	Root-leaf	9.7	1.1	1.3	8.5x	7.3x	20.5	0.8	0.9	24x	23x	1.5	1.1	1.1	1.3x	1.4x
DT100	<i>Qa&d</i>	Deep parent-leaf	10.2	1.4	1.3	7.5x	7.6x	21.8	0.9	0.9	23x	23x	1.2	1.1	1.2	1.1x	1.0x
DT100	<i>Qa&d</i>	Siblings	9.5	1.2	1.1	8.0x	8.3x	19.4	1.0	0.9	19x	22x	1.3	1.2	1.2	1.1x	1.0x
DT100	<i>Qa&d</i>	Distant leaves	9.6	1.2	1.1	8.1x	8.5x	18.8	0.9	0.9	20x	20x	1.4	1.3	0.7	1.1x	2.1x
DT1K	<i>Qa&d</i>	Root-leaf	11.0	1.4	1.3	7.6x	8.6x	20.3	0.9	1.0	24x	21x	1.8	1.5	1.2	1.3x	1.5x
DT1K	<i>Qa&d</i>	Deep parent-leaf	10.8	1.2	1.2	8.7x	9.0x	19.9	0.9	0.9	21x	21x	1.3	1.4	1.2	0.97x	1.1x
DT1K	<i>Qa&d</i>	Siblings	11.2	2.7	2.5	4.2x	4.4x	17.5	0.9	0.9	19x	19x	1.4	1.3	1.3	1.1x	1.0x
DT1K	<i>Qa&d</i>	Distant leaves	10.7	2.6	2.5	4.1x	4.2x	20.1	1.0	0.9	21x	21x	1.6	1.3	1.3	1.2x	1.2x
DT10K	<i>Qa&d</i>	Root-leaf	18.7	1.6	1.6	12x	12x	21.9	0.9	1.0	24x	21x	1.6	1.2	1.1	1.4x	1.5x
DT10K	<i>Qa&d</i>	Deep parent-leaf	18.0	1.5	1.6	12x	11x	20.1	0.9	0.9	21x	22x	1.3	1.1	1.1	1.2x	1.2x
DT10K	<i>Qa&d</i>	Siblings	21.5	16.8	16.7	1.3x	1.3x	19.8	0.9	1.0	23x	20x	1.4	1.2	1.2	1.1x	1.1x
DT10K	<i>Qa&d</i>	Distant leaves	21.2	16.8	16.7	1.3x	1.3x	20.3	1.0	1.0	21x	21x	1.6	1.2	1.2	1.3x	1.3x
DT100K	<i>Qa&d</i>	Root-leaf	80.0	1.6	1.6	49x	50x	19.7	1.0	1.0	19x	19x	1.1	1.0	0.8	1.1x	1.3x
DT100K	<i>Qa&d</i>	Deep parent-leaf	80.8	1.4	1.3	60x	60x	21.3	1.0	0.9	21x	23x	0.9	0.8	1.1	1.0x	0.75x
DT100K	<i>Qa&d</i>	Siblings	130.8	160.3	155.3	0.82x	0.84x	20.0	1.0	1.0	21x	21x	1.2	1.2	1.3	0.94x	0.92x
DT100K	<i>Qa&d</i>	Distant leaves	130.2	162.2	156.9	0.80x	0.83x	21.1	0.9	0.9	25x	23x	1.4	1.3	1.3	1.1x	1.1x

Queries: *Qdesc* = All Descendants; *Qchild* = All Children; *Qleaf* = All Leaves; *Qa&d* = Check if Ancestor.
Parameters: Root-leaf = root-farthest-leaf pair; Deep parent-leaf = deepest parent-leaf pair (both true).
Siblings = shallow sibling pair; Distant leaves = leaf pair with greatest distance via their lowest common ancestor (both false).
Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

Check if Ancestor (Query 11)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
WT10	<i>Qa&d</i>	Root-leaf	10.0	1.3	1.3	7.5x	7.8x	18.4	0.9	0.9	20x	21x	1.8	1.1	1.1	1.6x	1.6x
WT10	<i>Qa&d</i>	Deep parent-leaf	10.0	1.3	1.3	7.5x	7.6x	18.9	0.8	1.0	22x	20x	1.3	1.2	1.2	1.1x	1.1x
WT10	<i>Qa&d</i>	Siblings	9.4	1.0	1.0	9.1x	9.6x	18.5	0.9	0.9	21x	20x	1.4	1.3	1.4	1.0x	1.0x
WT10	<i>Qa&d</i>	Distant leaves	9.7	1.0	1.0	9.8x	10x	17.6	0.9	0.9	19x	19x	1.5	1.3	1.3	1.1x	1.2x
WT100	<i>Qa&d</i>	Root-leaf	11.4	1.7	1.6	6.7x	7.1x	19.1	0.9	0.9	21x	20x	1.4	1.0	1.1	1.4x	1.3x
WT100	<i>Qa&d</i>	Deep parent-leaf	11.1	1.5	1.5	7.5x	7.6x	18.3	0.9	0.9	20x	19x	1.2	1.1	1.3	1.1x	0.98x
WT100	<i>Qa&d</i>	Siblings	9.8	1.2	1.3	8.3x	7.7x	17.6	0.9	0.9	19x	19x	1.3	1.1	1.1	1.1x	1.2x
WT100	<i>Qa&d</i>	Distant leaves	9.8	1.1	1.1	8.6x	8.8x	21.2	0.9	1.0	23x	22x	1.7	1.4	1.4	1.2x	1.2x
WT1K	<i>Qa&d</i>	Root-leaf	12.7	1.6	1.5	8.1x	8.3x	20.7	0.8	0.9	25x	23x	1.1	0.8	1.1	1.3x	0.94x
WT1K	<i>Qa&d</i>	Deep parent-leaf	12.1	1.5	1.5	7.8x	8.0x	18.6	0.8	0.9	22x	21x	1.3	1.3	1.3	1.0x	0.99x
WT1K	<i>Qa&d</i>	Siblings	10.8	2.6	2.5	4.2x	4.3x	21.0	0.9	1.1	24x	19x	1.4	1.4	1.4	1.1x	1.0x
WT1K	<i>Qa&d</i>	Distant leaves	10.6	2.6	2.5	4.1x	4.2x	22.6	0.9	0.9	24x	25x	1.6	1.4	1.4	1.1x	1.2x
WT10K	<i>Qa&d</i>	Root-leaf	19.1	1.7	1.8	11x	11x	21.2	0.9	0.9	25x	25x	1.4	1.2	0.9	1.2x	1.5x
WT10K	<i>Qa&d</i>	Deep parent-leaf	18.5	1.8	1.7	11x	11x	21.0	1.0	0.9	22x	23x	1.2	1.1	1.1	1.1x	1.1x
WT10K	<i>Qa&d</i>	Siblings	20.5	16.4	16.2	1.2x	1.3x	19.3	0.9	0.8	22x	23x	1.3	1.2	1.2	1.1x	1.1x
WT10K	<i>Qa&d</i>	Distant leaves	20.9	16.6	16.4	1.3x	1.3x	20.8	0.8	0.8	25x	25x	1.4	1.2	1.2	1.2x	1.1x
WT100K	<i>Qa&d</i>	Root-leaf	79.8	1.9	1.8	42x	44x	20.2	0.9	0.9	22x	23x	1.6	1.3	1.3	1.2x	1.2x
WT100K	<i>Qa&d</i>	Deep parent-leaf	81.2	1.8	1.8	44x	45x	22.5	0.8	0.9	27x	25x	1.3	1.3	1.3	1.0x	0.96x
WT100K	<i>Qa&d</i>	Siblings	126.2	162.5	157.3	0.78x	0.80x	20.0	0.9	0.9	23x	22x	1.2	1.3	1.4	0.96x	0.89x
WT100K	<i>Qa&d</i>	Distant leaves	125.9	160.4	159.4	0.79x	0.79x	19.6	0.9	0.9	22x	21x	1.4	1.4	1.5	1.0x	0.98x
SNB/C	<i>Qa&d</i>	Root-leaf	14,648.6	2.0	1.9	7.3Kx	7.9Kx	22.3	1.0	0.9	22x	24x	1.6	1.2	1.1	1.3x	1.4x
SNB/C	<i>Qa&d</i>	Deep parent-leaf	14,582.5	2.1	1.8	6.9Kx	8.0Kx	19.4	1.3	1.0	15x	19x	1.2	1.1	1.1	1.1x	1.0x
SNB/C	<i>Qa&d</i>	Siblings	15,882.2	3,576.6	3,516.9	4.4x	4.5x	18.7	1.1	0.9	18x	22x	1.2	1.1	1.2	1.0x	1.0x
SNB/C	<i>Qa&d</i>	Distant leaves	15,854.3	3,769.2	3,664.0	4.2x	4.3x	20.3	1.1	0.8	19x	24x	1.2	1.1	1.1	1.1x	1.1x
SNB/P	<i>Qa&d</i>	Root-leaf	14,525.6	1.1	1.5	13Kx	9.7Kx	19.6	0.8	0.9	24x	22x	1.4	1.1	1.1	1.3x	1.3x
SNB/P	<i>Qa&d</i>	Deep parent-leaf	14,422.9	1.6	1.5	9.2Kx	9.7Kx	17.3	0.9	0.9	20x	19x	1.3	1.1	1.1	1.2x	1.1x
SNB/P	<i>Qa&d</i>	Siblings	14,476.3	3.2	3.2	4.5Kx	4.6Kx	20.9	0.9	0.9	24x	23x	1.2	1.1	1.2	1.1x	1.0x
SNB/P	<i>Qa&d</i>	Distant leaves	14,341.4	3.2	3.1	4.5Kx	4.6Kx	17.6	0.9	0.9	20x	20x	1.3	1.2	1.2	1.1x	1.1x
SNB/T	<i>Qa&d</i>	Root-leaf	14,538.5	1.6	1.7	9.0Kx	8.6Kx	18.9	0.8	0.9	23x	20x	1.2	1.1	1.2	1.1x	1.0x
SNB/T	<i>Qa&d</i>	Deep parent-leaf	14,437.5	1.4	1.7	10Kx	8.6Kx	20.5	0.8	0.9	25x	23x	1.4	1.1	1.1	1.3x	1.3x
SNB/T	<i>Qa&d</i>	Siblings	14,251.4	1.1	1.0	13Kx	14Kx	19.3	0.8	0.9	23x	22x	1.2	1.1	1.2	1.1x	1.0x
SNB/T	<i>Qa&d</i>	Distant leaves	14,278.8	1.0	1.0	14Kx	14Kx	18.3	0.8	0.8	22x	22x	1.3	1.5	1.2	0.87x	1.1x

Queries: *Qdesc* = All Descendants; *Qchild* = All Children; *Qleaf* = All Leaves; *Qa&d* = Check if Ancestor.

Parameters: Root-leaf = root-farthest-leaf pair; Deep parent-leaf = deepest parent-leaf pair (both true).

Siblings = shallow sibling pair; Distant leaves = leaf pair with greatest distance via their lowest common ancestor (both false).

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

LDBC SNB SF1: IC12, IS2, IS6

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
SNB	IC12	-	188,529.8	170,910.6	173,980.7	1.1x	1.1x	396.5	304.9	403.9	1.3x	0.98x	157.5	119.4	106.3	1.3x	1.5x
SNB	IS2	-	61,383.9	3,842.9	13,060.7	16x	4.7x	43.7	242.2	277.4	0.18x	0.16x	4.5	4.5	10.7	0.99x	0.42x
SNB	IS6	-	17,296.9	2,172.2	3,269.6	8.0x	5.3x	28.8	22.9	118.4	1.3x	0.24x	1.9	2.0	1.8	0.95x	1.0x

Queries: IC12 = Interactive Complex 12; IS2 = Interactive Short 2; IS6 = Interactive Short 6.

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. Runtimes: median of five runs in ms.

Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.

SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes to one decimal; speedups to about two significant digits; K = 1,000x. Bounds round outward.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup